

8C

QUANTUM MODEL OF ATOM

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 8

1. Describe and apply the dual nature of light.
2. Calculate energy, frequency, or wavelength of a sample of light.
3. Use the Bohr model of energy levels in atoms to explain light emission and absorption by gaseous atoms.
4. Calculate the energy or wavelength of a specific electronic transition in a hydrogen atom.
5. Visualize and describe the spatial distributions and energies of atomic electrons using the quantum model of the atom.
6. Use the concept of quantum numbers to characterize the energies, spatial distributions, and spins of atomic electrons.
7. Sketch the electronic energy levels of an atom.
8. Add electrons to atomic orbital energy (energy level) diagrams.
9. Determine electron configurations for the elements using the concepts of shells, subshells, and orbitals.

Shapes of Orbitals

8.5

Relative sizes and energies of orbitals:

- Size and energy generally increase with increasing shell (n)
- Within a shell (n), the energy of subshells is: $E_s < E_p < E_d < E_f$

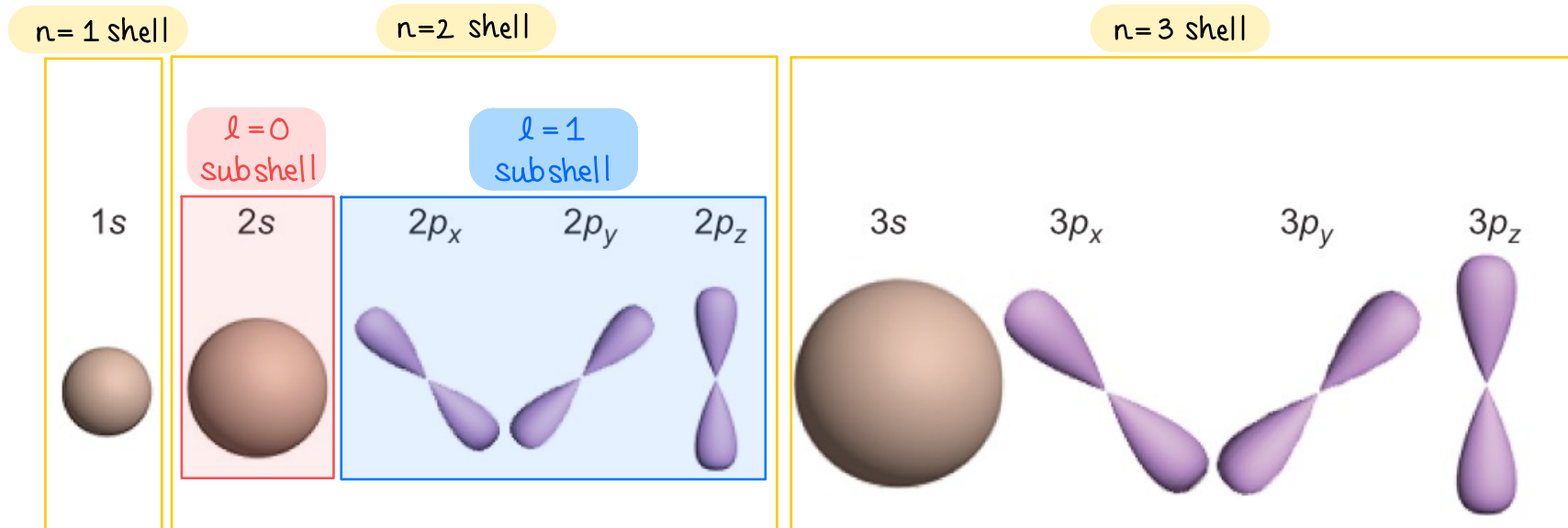
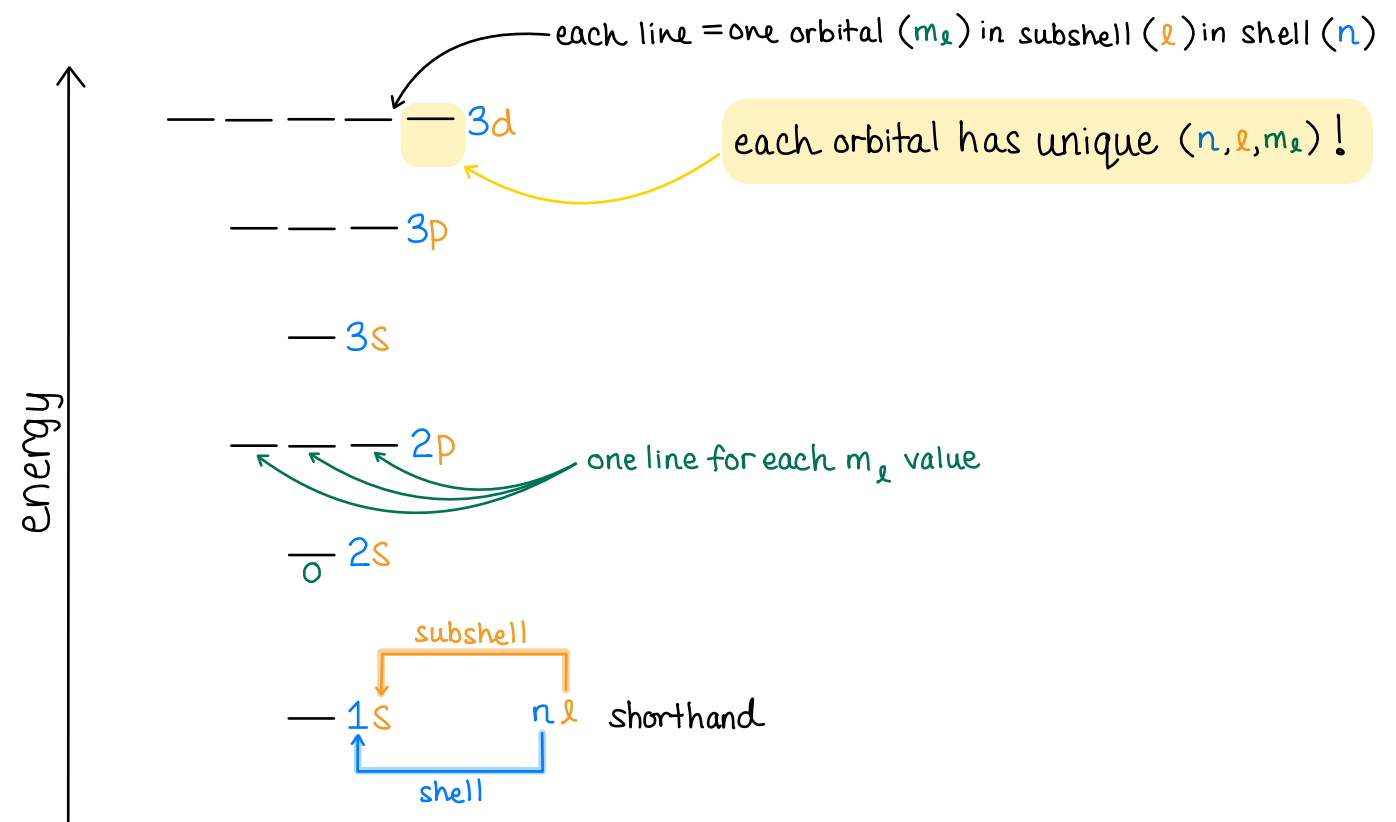
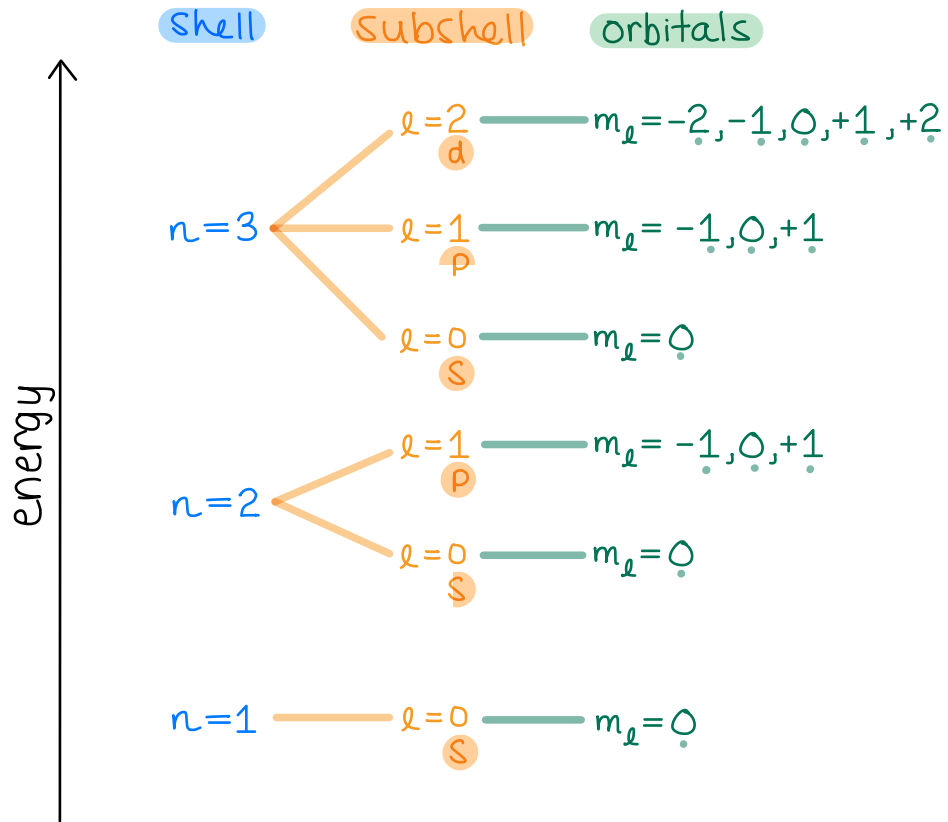


FIGURE 8.18

Putting it all together

Chemistry in Context

No two orbitals and no two electrons have the same set of quantum numbers.



IN-CLASS QUESTION 1

Learning Objective(s): 8.5, 8.6

Which set of quantum numbers could be associated with an electron that exists in the orbital shown below?

☐ ~~$n = 2, \ell = 2, m_\ell = -1, m_s = +\frac{1}{2}$~~

$\ell \neq 2$ if $n=2$; i.e. 2d not possible!

☐ $n = 4, \ell = 3, m_\ell = +2, m_s = -\frac{1}{2}$

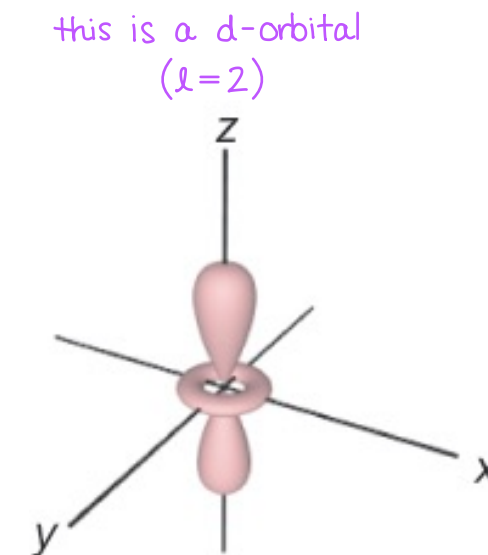
this would be an f-orbital

☐ $n = 4, \ell = 2, m_\ell = +3, m_s = +\frac{1}{2}$

$m_\ell \neq +3$ if $\ell=2$

☒ $n = 3, \ell = 2, m_\ell = 0, m_s = +\frac{1}{2}$

this would be a 3d-orbital ($\ell=2$)



Orbital Diagrams

8.6

Orbital energy-level diagrams

- Orbitals within a **subshell** (s, p, d, f) are **degenerate**: same energy
- The ordering of orbital energies can be determined from the periodic table.

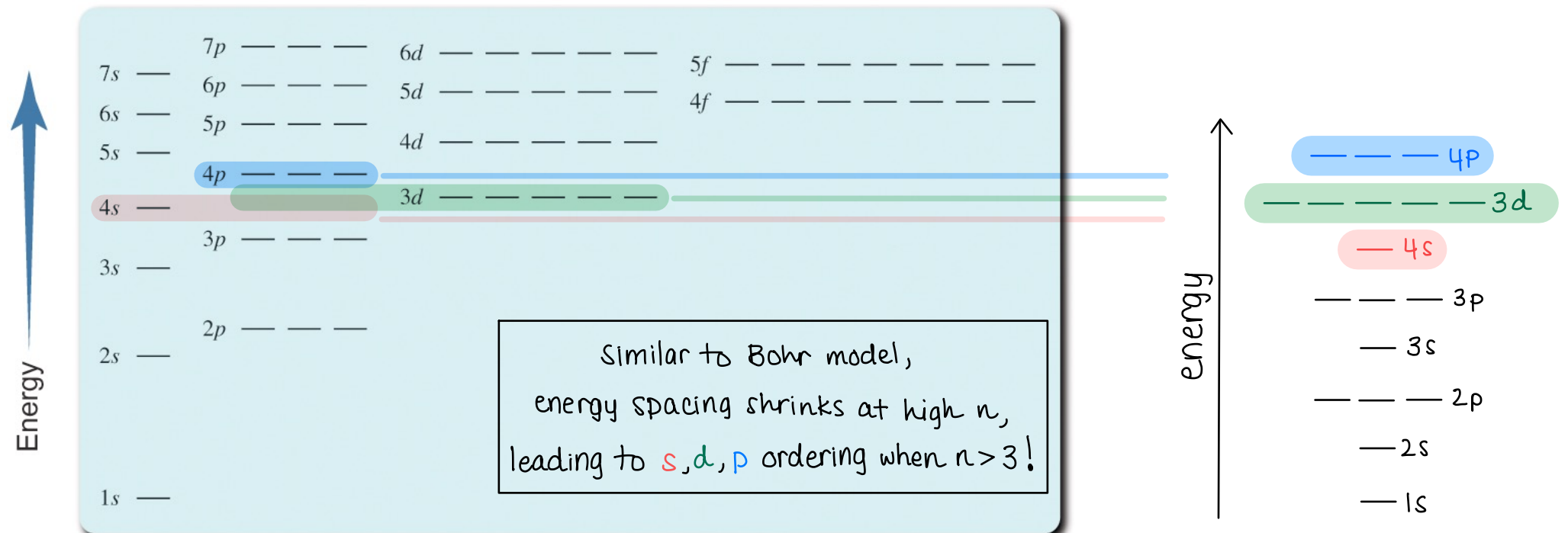


FIGURE 8.20

Art Conservation

X-ray fluorescence (XRF) on Jacques Louis David's
Antoine Laurent and Marie Anne Lavoisier at The MET.

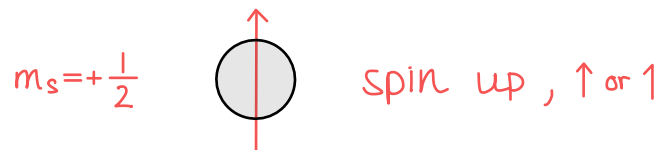


Orbital Diagrams

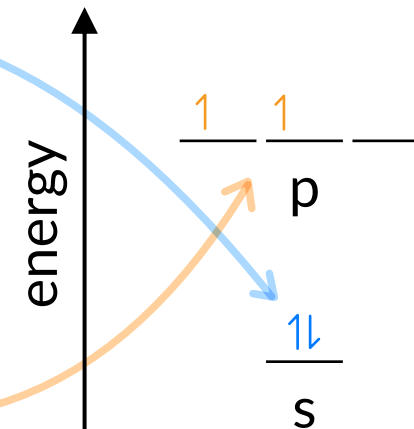
8.6

Pauli Exclusion Principle: no two electrons share the same set of quantum numbers.

- Electrons are represented by **arrows pointing up or down** that correlate to the two possible “spin” orientations denoted by the two spin quantum numbers:



- If one **orbital** has two electrons, then the electrons must have **OPPOSITE** spins (i.e., **OPPOSITE** values of m_s).
- If one **subshell** has *unpaired electrons*, then the electrons must have **THE SAME** spins (i.e., **THE SAME** values of m_s).

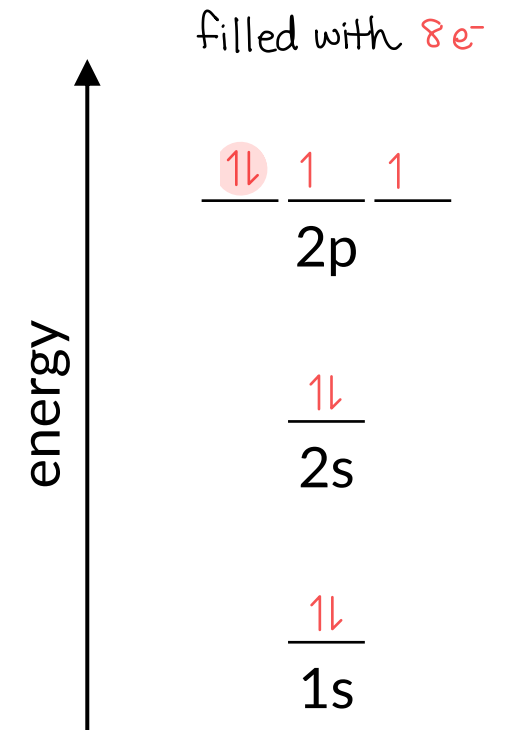
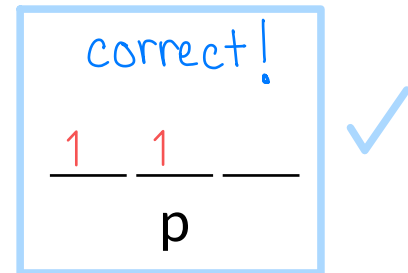
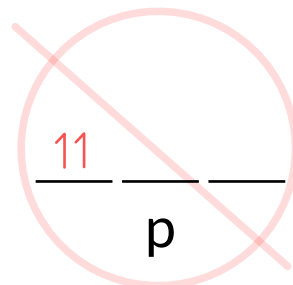
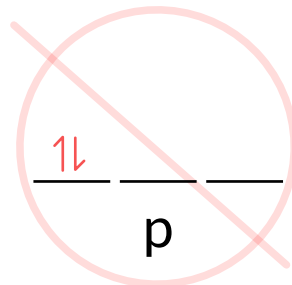


Orbital Diagrams

8.6

How to fill or occupy the orbital energy-level diagrams for an atom?

- Aufbau principle (*building-up principle*): fill bottom to top
 - Fill electrons into orbitals from lowest to highest in energy.
- Hund's rule: fill across before pairing
 - Within a subshell (ℓ), avoid pairing electrons until necessary by *maximizing unpaired electron spins* across degenerate orbitals.
 - For example, 2 electrons in p subshell:

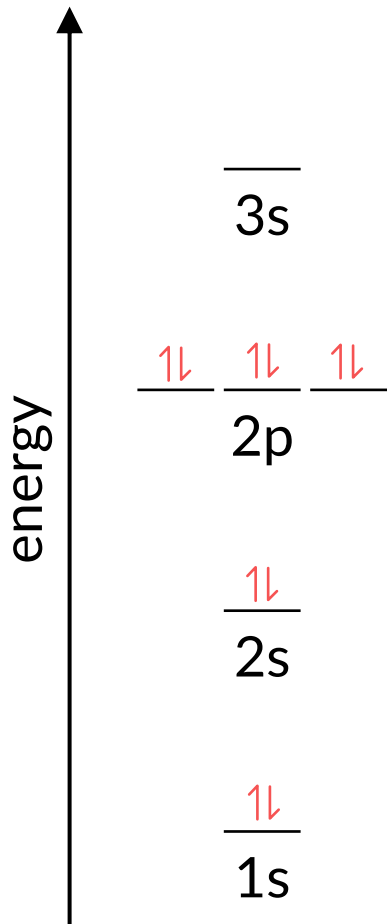


Electron Configurations

8.7

orbital
energy
diagram

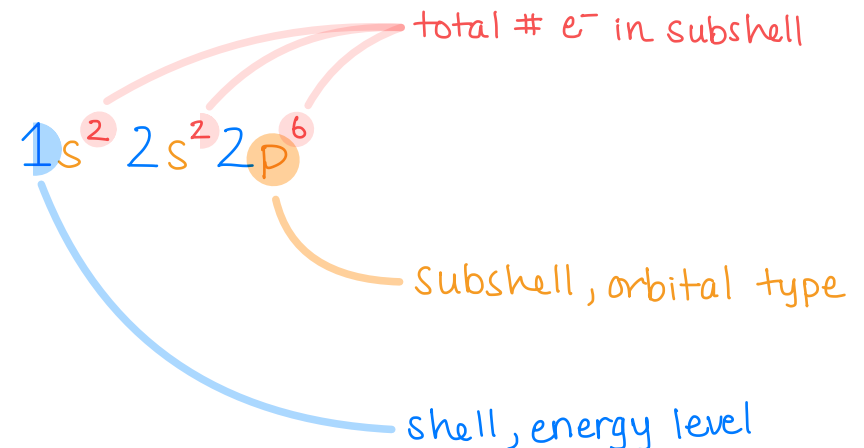
Ne



An electron configuration for an atom shows the particular orbitals that electrons occupy for that atom.

- Ground state electron configuration: the lowest energy state or configuration for all electrons in an atom.
- For example, the electron configuration for neon (Ne) is

Ne ($Z=10$) , $10e^-$



Electron Configurations

8.7

For any element, its highest-energy shell (n) is given by its position on the periodic table.

s-block elements	p-block elements	d-block elements	f-block elements
$n = (\text{row number})$		$n = (\text{row number}) - 1$	$n = (\text{row number}) - 2$

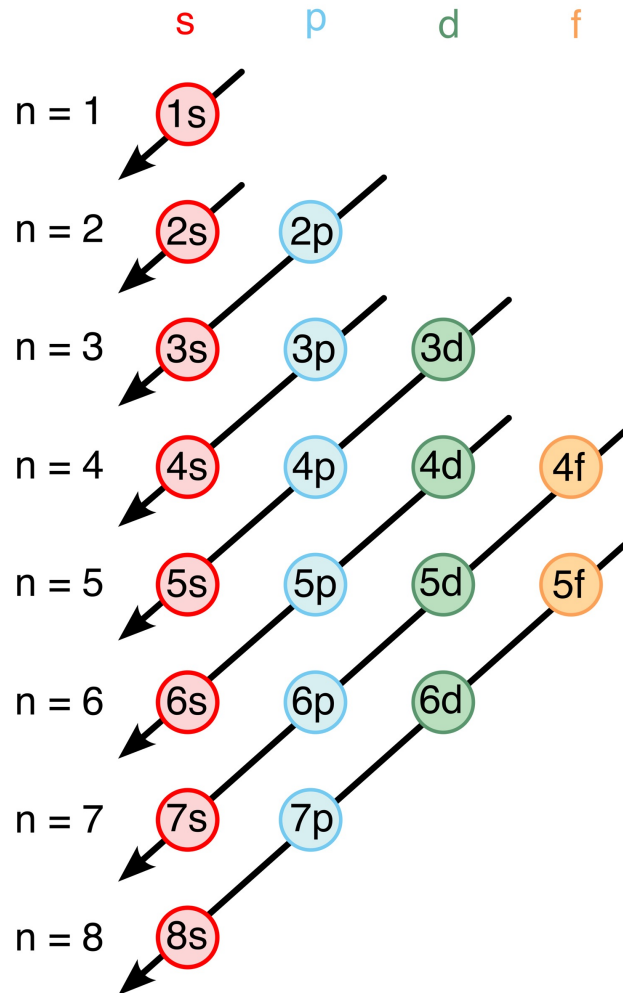
H:	$Z=1$	$1s^1$
He:	$Z=2$	$1s^2$
Li:	$Z=3$	$1s^2 2s^1$
Be:	$Z=4$	$1s^2 2s^2$
B:	$Z=5$	$1s^2 2s^2 2p^1$
N:	$Z=7$	$1s^2 2s^2 2p^3$
Cl:	$Z=17$	$1s^2 2s^2 2p^6 3s^2 3p^5$
Ca:	$Z=20$	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
Sc:	$Z=21$	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$

s-block												p-block					
1	2											13	14	15	16	17	18
1	1s																1s
2	2s	d-block												2p			
3	3s	3	4	5	6	7	8	9	10	11	12			3p			
4	4s					3d								4p			
5	5s					4d								5p			
6	6s	*				5d								6p			
7	7s	†				6d								7p			
		f-block															
6	*							4f									
7	†							5f									

FIGURE 8.25

“Periodic” Table of Elements

Chemistry in Context



The filling/placement of electrons into orbitals is systematic. You can use the “triangular method” (left image) or rely on the organization of the periodic table into blocks (below).

	S-block		p-block												
	1	2													18
1	1s														1s
2	2s		d-block											2p	
3	3s		3	4	5	6	7	8	9	10	11	12		3p	
4	4s							3d						4p	
5	5s							4d						5p	
6	6s		*					5d						6p	
7	7s		†					6d						7p	
f-block															
6	*									4f					
7	†									5f					

FIGURE 8.25

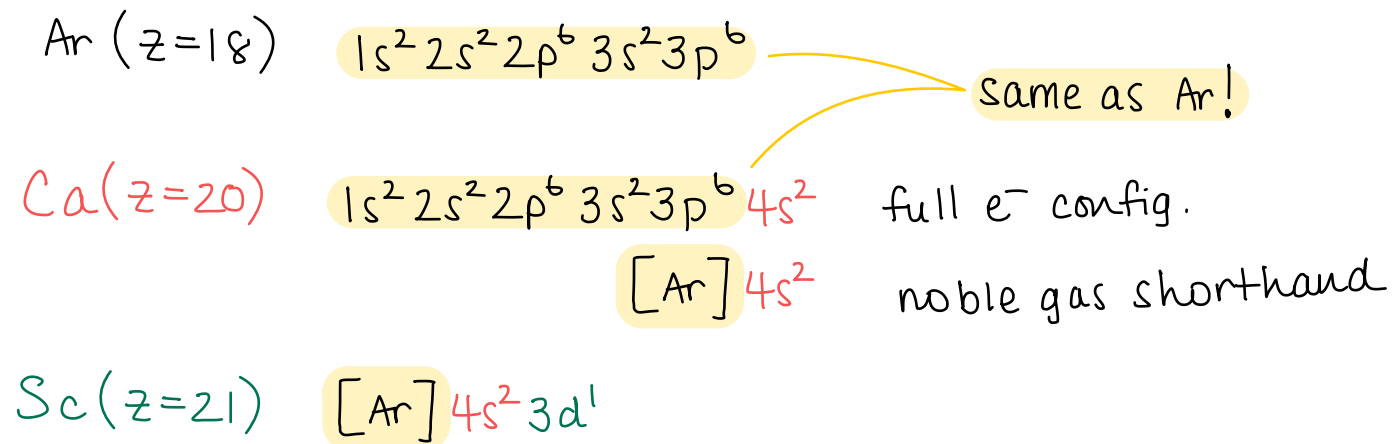
Electron Configurations

8.7

Abbreviated noble-gas electron configuration

- Shorthand based on the idea that any atom contains all the same electrons as the noble gas immediately *preceding* it.

Example: Write the abbreviated electron configurations for calcium and scandium.

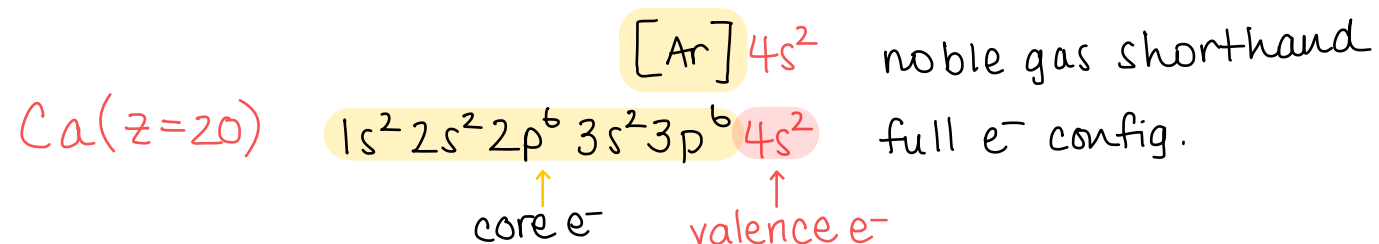


Electron Configurations

8.7

Core electrons

- The “inner” electrons, which are lower in energy and lie closer to the nucleus.
- Usually, these are all the electrons in prior rows (i.e., the filled shells and subshells) represented by the noble gas shorthand.



Valence electrons

- The “outermost” (highest in energy) electrons shown explicitly when a noble gas shorthand is used to write electronic configurations.
- For **s-block** and **p-block** elements: e^- in outermost shell (highest n)

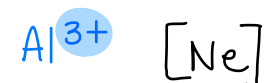
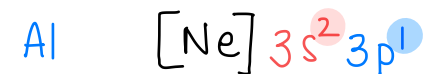
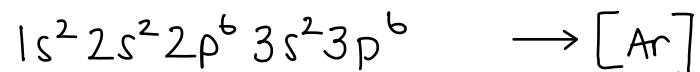
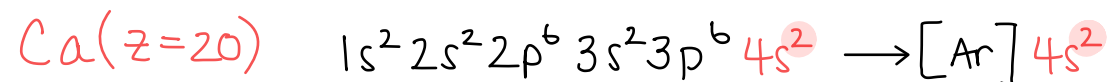
Electron Configurations

8.7

Cations

- Form when an atom *loses* one or more electrons.
- The most common monatomic cations of the **s-block** and **p-block** elements have noble gas electron configuration.
- All atoms lose their *highest energy electrons first* to form cations.
 - For **s-block** and **p-block** elements, these are the valence electrons in the s- and p-orbitals of the highest value of n .

Example: Write the electron configurations for the calcium and aluminum cations.



Cr

$Z = 24$
neutral

Electron Configurations

8.7

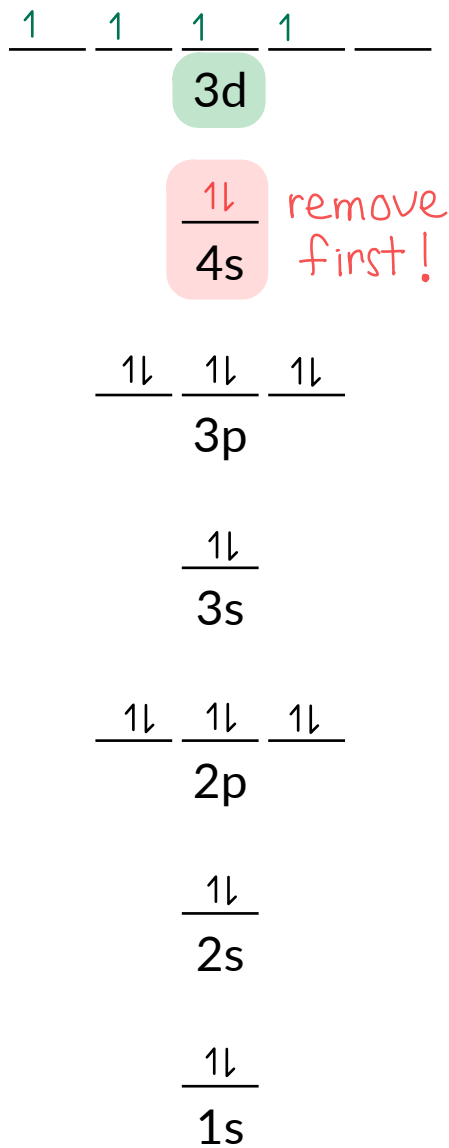
EXCEPTION: Cations of transition metal (**d-block**) elements

- First two electrons lost from the highest energy **s-orbital**!
- Subsequent electrons are then lost from the **d-orbitals**.

Example: Write the electron configurations for Cr, Cr⁺, Cr²⁺, and Cr³⁺.



energy



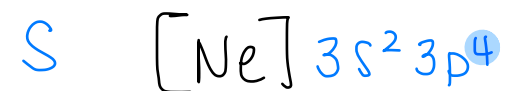
Electron Configurations

8.7

Anions

- Formed when an atom *gains* one or more electrons.
- The most common monatomic anions of **s-block** and **p-block** elements have noble gas electron configuration.

Example: Write the electron configuration for the sulfide anion.



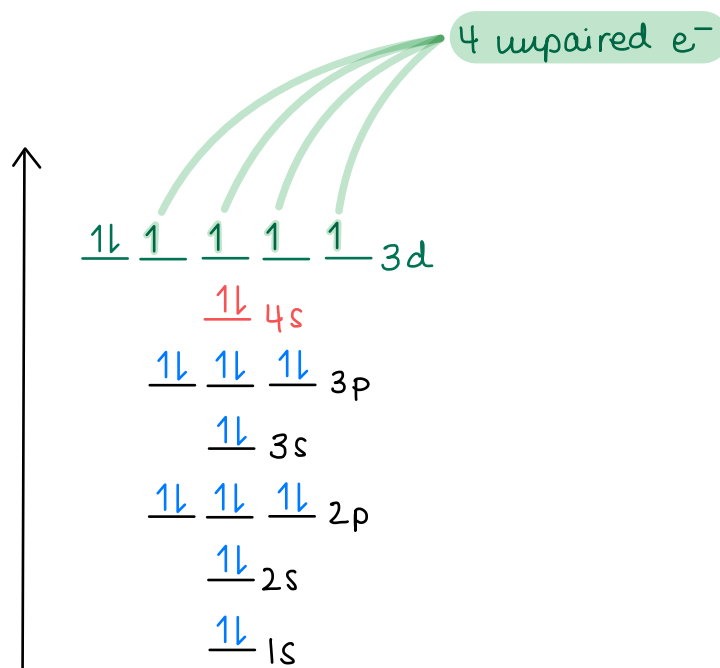
IN-CLASS QUESTION 2

Learning Objective(s): 8.8, 8.9

How many *unpaired* electrons are in the ground-state electron configuration for a neutral iron atom?

Fe, $z = 26$, $26 e^-$

Fe : $[Ar] 4s^2 3d^6$



IN-CLASS QUESTION 3

Learning Objective(s): 8.8, 8.9

How many *unpaired* electrons are in the ground-state electron configuration for the Fe^{3+} cation?

For transition metal cations:

— remove $ns\ e^-$ first

— then remove $(n-1)d\ e^-$ as needed

Fe , $z = 26$, $26\ e^-$

Fe^{3+} , $z = 26$, $23\ e^-$

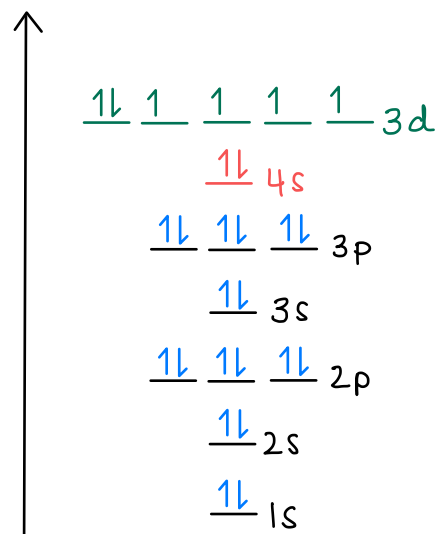
Fe : $[\text{Ar}] 4s^2 3d^6$

Fe^+ : $[\text{Ar}] 4s^1 3d^6$

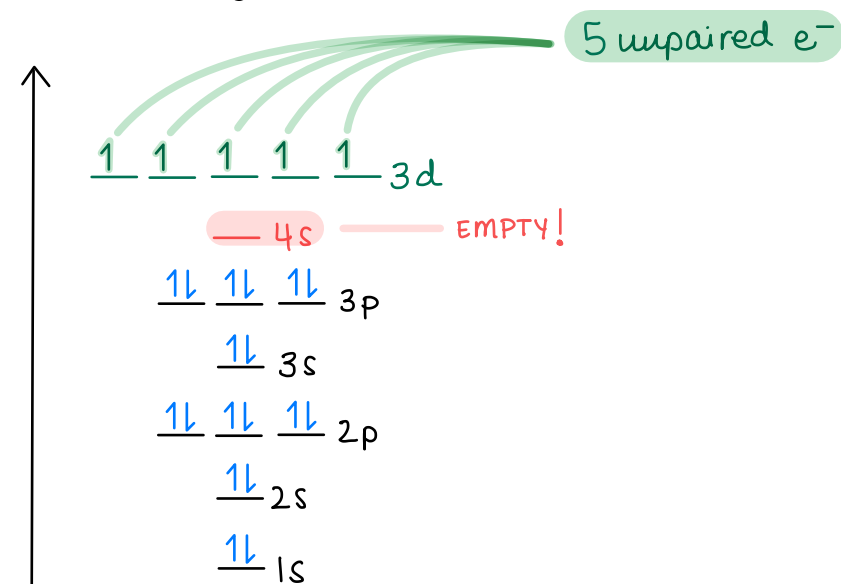
Fe^{2+} : $[\text{Ar}] 3d^6$

Fe^{3+} : $[\text{Ar}] 3d^5$

neutral Fe



cation Fe^{3+}



BONUS QUESTION

Assessed on: Exam 2, Fall 2021

Write the chemical symbol (e.g., He for helium) for each element based on the description of its ground state electron configuration.

Chemical Symbol	Description of Gound State Electron Configuration
	This element has four valence electrons with two unpaired electrons in the $4d$ orbitals.
	This element has five valence electrons with three unpaired electrons in the $3p$ orbitals.
	This element has six valence electrons with two unpaired electrons in the $4p$ orbitals.